MODEL LAB · 2026-10-15

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

26,804
held-out plate appearances

3.05%
held-out HR rate

0.6058
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-03, scored on the 26,804 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5778	1.66×	0.02953	0.13565	0.00621
KNN comps Prior is the pooled rate of the nearest comparables.	0.6058	1.55×	0.02947	0.13432	0.00421
SVR Support-vector regression on the same feature matrix.	0.5969	1.53×	0.02952	0.13512	0.00608
Random forest Forest regression; a learned similarity metric.	0.5978	1.53×	0.02951	0.13473	0.00580
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6058	1.55×	0.02947	0.13432	0.00421
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6058	1.55×	0.02947	0.13432	0.00421
Linear regression Ordinary least squares on the same features.	0.6003	1.54×	0.02951	0.13478	0.00444
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6008	1.64×	0.02953	0.13487	0.00658
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6001	1.59×	0.02950	0.13464	0.00483
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5977	1.54×	0.02954	0.13496	0.00695

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,077 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5874	1.25×	0.02287	0.11127	0.01213
KNN comps Prior is the pooled rate of the nearest comparables.	0.6175	1.94×	0.02281	0.10947	0.01009
SVR Support-vector regression on the same feature matrix.	0.5928	2.08×	0.02307	0.11387	0.01831
Random forest Forest regression; a learned similarity metric.	0.6057	2.08×	0.02302	0.11169	0.01872
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6175	1.94×	0.02281	0.10947	0.01009
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6175	1.94×	0.02281	0.10947	0.01009
Linear regression Ordinary least squares on the same features.	0.6136	1.94×	0.02308	0.11210	0.01708
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6205	2.22×	0.02312	0.11132	0.01760
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6203	2.22×	0.02286	0.10990	0.01274
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6098	1.94×	0.02329	0.11305	0.02431

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 16.5%, Jackson Chourio 16.2%, Jonathan Aranda 15.4%, Junior Caminero 15.0%, Jake Bauers 14.6%, William Contreras 14.6%
Flat league	6/6	Jackson Chourio 16.3%, Shohei Ohtani 14.9%, Jonathan Aranda 14.6%, Junior Caminero 14.5%, William Contreras 14.4%, Jake Bauers 13.7%
KNN comps	6/6	Jackson Chourio 16.5%, Shohei Ohtani 16.4%, Jonathan Aranda 15.6%, Junior Caminero 14.9%, Jake Bauers 14.9%, William Contreras 14.6%
SVR	5/6	Jackson Chourio 17.0%, Shohei Ohtani 16.3%, Junior Caminero 16.2%, Jonathan Aranda 15.0%, Jake Bauers 15.0%, Victor Mesa Jr. 14.6%
Random forest	6/6	Jackson Chourio 17.1%, Shohei Ohtani 16.6%, Junior Caminero 15.3%, Jonathan Aranda 15.2%, Jake Bauers 15.1%, William Contreras 14.4%
Core (KNN+SVR)	6/6	Jackson Chourio 16.5%, Shohei Ohtani 16.4%, Jonathan Aranda 15.6%, Junior Caminero 14.9%, Jake Bauers 14.9%, William Contreras 14.6%
Forest blend	6/6	Jackson Chourio 16.5%, Shohei Ohtani 16.4%, Jonathan Aranda 15.6%, Junior Caminero 14.9%, Jake Bauers 14.9%, William Contreras 14.6%
Linear regression	5/6	Jackson Chourio 16.8%, Shohei Ohtani 16.3%, Jonathan Aranda 15.3%, Junior Caminero 15.1%, Jake Bauers 14.7%, Will Smith 14.5%
Logistic regression	6/6	Jackson Chourio 16.8%, Shohei Ohtani 16.7%, Junior Caminero 15.5%, Jonathan Aranda 15.1%, Jake Bauers 14.7%, William Contreras 14.5%
Neural net (torch)	6/6	Jackson Chourio 16.9%, Shohei Ohtani 16.6%, Junior Caminero 15.5%, Jonathan Aranda 15.2%, Jake Bauers 14.8%, William Contreras 14.5%
XGBoost	6/6	Jackson Chourio 17.3%, Shohei Ohtani 17.0%, Junior Caminero 15.8%, Jonathan Aranda 15.4%, Jake Bauers 15.1%, William Contreras 14.6%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Shohei Ohtani LAD slate	**16.5%**	14.9%	16.4%	16.3%	16.6%	16.4%	16.4%	16.3%	16.7%	
2 Jackson Chourio MIL slate	**16.2%**	16.3%	16.5%	17.0%	17.1%	16.5%	16.5%	16.8%	16.8%	
3 Jonathan Aranda TB slate	**15.4%**	14.6%	15.6%	15.0%	15.2%	15.6%	15.6%	15.3%	15.1%	
4 Junior Caminero TB	**15.0%**	14.5%	14.9%	16.2%	15.3%	14.9%	14.9%	15.1%	15.5%	
5 Jake Bauers MIL	**14.6%**	13.7%	14.9%	15.0%	15.1%	14.9%	14.9%	14.7%	14.7%	
6 William Contreras MIL	**14.6%**	14.4%	14.6%	14.4%	14.4%	14.6%	14.6%	14.5%	14.5%	
7 Will Smith LAD	**13.9%**	13.6%	13.8%	13.8%	14.3%	13.8%	13.8%	14.5%	14.3%	
8 Teoscar Hernández LAD	**13.9%**	13.2%	14.1%	13.9%	14.1%	14.1%	14.1%	14.0%	14.1%	
9 Jo Adell CLE slate	**13.8%**	13.1%	13.5%	14.0%	13.6%	13.5%	13.5%	13.7%	14.1%	
10 Victor Mesa Jr. TB	**13.8%**	13.6%	14.1%	14.6%	14.3%	14.1%	14.1%	14.2%	14.1%	
11 Patrick Bailey CLE	**13.4%**	12.7%	13.4%	13.2%	12.9%	13.4%	13.4%	13.2%	12.8%	
12 Mookie Betts LAD	**13.2%**	13.1%	13.0%	13.4%	13.1%	13.0%	13.0%	13.2%	13.2%	
13 Brice Turang MIL	**12.9%**	13.2%	12.8%	12.8%	12.7%	12.8%	12.8%	12.3%	12.2%	
14 Freddie Freeman LAD	**12.7%**	12.4%	12.5%	12.3%	12.5%	12.5%	12.5%	12.7%	12.5%	
15 Chase DeLauter CLE	**12.3%**	12.3%	12.2%	12.0%	12.1%	12.2%	12.2%	12.1%	11.9%	
16 Yandy Díaz TB	**11.8%**	12.2%	11.9%	12.4%	12.2%	11.9%	11.9%	12.0%	11.9%	
17 Christian Yelich MIL	**11.8%**	11.9%	12.0%	11.6%	12.0%	12.0%	12.0%	12.1%	11.7%	
18 Garrett Mitchell MIL	**11.7%**	11.2%	11.6%	10.5%	10.9%	11.6%	11.6%	11.0%	10.9%	

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
19	**Travis Bazzana**	CLE	**11.7%**	11.7%	11.5%	11.4%	11.5%	11.5%	11.5%	11.6%	11.3%	
20	**Nathaniel Lowe**	CLE	**11.5%**	11.3%	11.7%	11.4%	11.7%	11.7%	11.7%	11.5%	11.4%	
21	**Kyle Tucker**	LAD	**10.9%**	11.0%	10.9%	10.6%	10.8%	10.9%	10.9%	10.8%	10.5%	
22	**Enrique Hernández**	LAD	**10.6%**	10.5%	10.7%	10.4%	10.6%	10.7%	10.7%	10.4%	10.1%	
23	**José Ramírez**	CLE	**10.6%**	10.9%	10.7%	10.8%	10.8%	10.7%	10.7%	11.0%	11.2%	
24	**Andy Pages**	LAD	**10.3%**	9.9%	10.0%	9.8%	10.0%	10.0%	10.0%	9.9%	9.7%	
25	**Petey Halpin**	CLE	**8.5%**	9.6%	8.6%	9.0%	8.8%	8.6%	8.6%	9.6%	9.0%	
26	**Miguel Rojas**	LAD	**7.5%**	9.2%	7.8%	7.8%	7.6%	7.8%	7.8%	8.1%	8.3%	
27	**Liam Hicks**	TB	**7.3%**	8.4%	7.2%	8.1%	7.3%	7.2%	7.2%	7.7%	7.5%	
28	**Brayan Rocchio**	CLE	**7.3%**	7.8%	7.2%	7.3%	6.9%	7.2%	7.2%	7.0%	7.0%	
29	**Jonny DeLuca**	TB	**7.2%**	7.8%	7.2%	7.1%	6.7%	7.2%	7.2%	7.0%	7.3%	
30	**Richie Palacios**	TB	**7.0%**	8.0%	7.1%	6.8%	6.5%	7.1%	7.1%	6.9%	6.9%	
31	**Cooper Pratt**	MIL	**6.2%**	8.0%	6.2%	5.2%	5.9%	6.2%	6.2%	5.4%	5.6%	
32	**Sal Frelick**	MIL	**6.1%**	7.8%	6.3%	6.0%	5.9%	6.3%	6.3%	6.0%	6.1%	
33	**David Hamilton**	MIL	**5.8%**	7.3%	5.7%	5.5%	5.3%	5.7%	5.7%	5.7%	5.6%	
34	**Steven Kwan**	CLE	**5.3%**	8.7%	5.7%	3.7%	4.3%	5.7%	5.7%	3.7%	5.3%	
35	**Taylor Walls**	TB	**4.8%**	6.6%	4.9%	4.1%	4.1%	4.9%	4.9%	5.0%	5.0%	
36	**Chandler Simpson**	TB	**2.9%**	6.3%	3.7%	2.3%	1.7%	3.7%	3.7%	1.7%	3.3%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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